

```
CREATE DATABASE sql_interview_db;
```

```
USE sql_interview_db;
```

```
CREATE TABLE departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50),  
    location VARCHAR(50)  
);
```

```
INSERT INTO departments VALUES  
(1, 'IT', 'Jaipur'),  
(2, 'HR', 'Delhi'),  
(3, 'Finance', 'Mumbai'),  
(4, 'Sales', 'Pune'),  
(5, 'Marketing', 'Jaipur');
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    gender VARCHAR(10),  
    department_id INT,  
    salary DECIMAL(10,2),  
    hire_date DATE,  
    manager_id INT,  
    city VARCHAR(50),  
    FOREIGN KEY (department_id) REFERENCES departments(dept_id)  
);
```

```
INSERT INTO employees VALUES
```

(101, 'Amit', 'Male', 1, 60000, '2021-01-15', NULL, 'Jaipur'),
(102, 'Ravi', 'Male', 1, 45000, '2022-03-20', 101, 'Jaipur'),
(103, 'Priya', 'Female', 2, 50000, '2020-07-10', NULL, 'Delhi'),
(104, 'Neha', 'Female', 2, 40000, '2023-02-15', 103, 'Delhi'),
(105, 'Rahul', 'Male', 3, 70000, '2019-11-25', NULL, 'Mumbai'),
(106, 'Pooja', 'Female', 3, 55000, '2021-06-18', 105, 'Mumbai'),
(107, 'Karan', 'Male', 4, 65000, '2020-09-12', NULL, 'Pune'),
(108, 'Anjali', 'Female', 4, 48000, '2022-12-01', 107, 'Pune'),
(109, 'Vikas', 'Male', 1, 75000, '2018-05-22', 101, 'Jaipur'),
(110, 'Simran', 'Female', 5, 52000, '2021-08-30', NULL, 'Jaipur'),
(111, 'Mohit', 'Male', 5, 38000, '2023-04-10', 110, 'Jaipur'),
(112, 'Sneha', 'Female', 4, 58000, '2020-01-05', 107, 'Pune'),
(113, 'Arjun', 'Male', 1, 62000, '2022-05-15', 101, 'Jaipur'),
(114, 'Kavita', 'Female', 3, 47000, '2023-07-20', 105, 'Mumbai'),
(115, 'Deepak', 'Male', 2, 43000, '2021-10-11', 103, 'Delhi');

```
CREATE TABLE projects (  
    project_id INT PRIMARY KEY,  
    project_name VARCHAR(100),  
    budget DECIMAL(12,2),  
    dept_id INT  
);
```

```
INSERT INTO projects VALUES  
(201, 'Website Development', 500000, 1),  
(202, 'Recruitment System', 200000, 2),  
(203, 'Financial Analysis', 800000, 3),  
(204, 'Sales Dashboard', 450000, 4),  
(205, 'Marketing Campaign', 300000, 5);
```

```
CREATE TABLE employee_projects (  
    emp_id INT,  
    project_id INT,  
    hours_worked INT,  
    PRIMARY KEY (emp_id, project_id),  
    FOREIGN KEY (emp_id) REFERENCES employees(emp_id),  
    FOREIGN KEY (project_id) REFERENCES projects(project_id)  
);
```

```
INSERT INTO employee_projects VALUES
```

```
(101, 201, 120),  
(102, 201, 100),  
(109, 201, 150),  
(113, 201, 90),  
(103, 202, 130),  
(104, 202, 110),  
(115, 202, 80),  
(105, 203, 160),  
(106, 203, 120),  
(114, 203, 100),  
(107, 204, 140),  
(108, 204, 100),  
(112, 204, 130),  
(110, 205, 120),  
(111, 205, 90);
```

```
select * from departments;
```

```
select * from employees;
```

```
select emp_name, salary , city from employees;
```

```
select city from employees where city ;
```

```
select emp_name,salary from employees where salary > 50000;
```

```
select emp_name,salary from employees where salary >= 50000;
```

```
select emp_name ,city from employees where city = 'jaipur';
```

```
select emp_name from employees where gender = 'female';
```

```
select emp_name,salary from employees where salary between 40000 and 60000;
```

```
select emp_name from employees where emp_name like'A%';
```

```
select emp_name from employees where emp_name like'%A';
```

```
select emp_name,salary from employees order by salary desc;
```

```
select emp_name from employees order by emp_name asc;
```

```
select emp_name ,salary from employees order by salary desc limit 5;
```

```
select emp_name ,salary from employees order by salary asc limit 3;
```

```
SELECT MAX(salary) AS highest_salary,MIN(salary) AS lowest_salary  
FROM employees;
```

```
select emp_name , hire_date from employees order by hire_date asc;
```

```
select count(*) from employees;
```

```
select sum(salary) from employees;
```

```
select avg(salary) from employees;
```

```
select max(salary) from employees;
```

```
select min(salary) from employees;
```

```
select count(gender) from employees where gender = 'male';
```

```
select count(gender) from employees where gender = 'female';
```

```
select avg(salary) from employees where city = 'jaipur';
```

```
select department_id ,count(*) from employees group by department_id;
```

```
select * from employees;
```

```
select department_id, avg(salary) from employees group by department_id;
```

```
select department_id, max(salary) from employees group by department_id;
```

```
select department_id, min(salary) from employees group by department_id;
```

```
select city, count(salary) from employees group by city;
```

```
select gender, count(salary) from employees group by gender;
```

```
select department_id, count(*) from employees group by department_id having  
count(*) >=2;
```

```
select department_id, avg(salary) from employees group by department_id having  
avg(salary) >50000;
```

```
select city, sum(salary) from employees group by city ;
```

```
select department_id, sum(salary) from employees group by department_id ;
```

```
select emp_name from employees e join department_id on e.emp_name =  
department_id.emp_name; -- error
```

```
show tables;
```

```
SELECT emp_name  
FROM employees;
```

```
SELECT emp_name, department  
FROM employees;
```

```
DESC employees;
```

```
SHOW COLUMNS FROM employees;
```

```
select emp_name from employees e join departments d ON e.department_id =  
d.dep_id;
```